Supplementary Information

A shared transcriptional response of *Mycobacterium marinum* to two microgravity simulators: a re-analysis of NASA OSDR study OSD-528

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Supplementary tables

All tables are tab-separated text files (one JSON file) in the `results_tables/` folder of the repository (`https://github.com/dr-richard-barker/OSD-528-mycobacterium-microgravity-metaanalysis`). Each is an output of the analysis pipeline, copied unchanged or joined from outputs; row counts are counted from the files.

Table S1 `Table_S1_kallisto_estimated_counts.tsv` (5,539 rows). kallisto estimated read-pair counts per CDS, all nine deposited samples (RFPRPM4 = inferred second library).

Table S2 `Table_S2_kallisto_tpm.tsv` (5,539 rows). Transcripts per million per CDS, all nine samples.

Table S3 `Table_S3_samples_and_sequencing_qc.tsv` (9 rows). Sample groups, primary/sensitivity use, library index, read pairs and read assignment (rRNA, other ncRNA, CDS).

Table S4 `Table_S4_deseq2_clinostat_vs_static1g.tsv` (3,968 rows). DESeq2: 3D clinostat vs static 1g (primary samples); MLE and apeglm-shrunk log2FC, Wald p, BH FDR.

Table S5 `Table_S5_deseq2_rpm2_vs_static1g.tsv` (3,968 rows). DESeq2: RPM 2.0 vs static 1g (primary samples).

Table S6 `Table_S6_deseq2_clinostat_vs_rpm2.tsv` (3,968 rows). DESeq2: 3D clinostat vs RPM 2.0 (primary samples).

Table S7 `Table_S7_shared_simulated_microgravity_genes.tsv` (533 rows). Genes differentially expressed under both simulators in the same direction.

Table S8 `Table_S8_deseq2_sensitivity_with_rfprpm4.tsv` (11,910 rows). DESeq2 results for all three contrasts with the inferred RFPRPM4 library included (sensitivity analysis).

Table S9 `Table_S9_wgcna_module_assignments.tsv` (3,968 rows). WGCNA module, module membership (kME) and connectivity (kTotal, kWithin) per gene.

Table S10 `Table_S10_wgcna_module_eigengenes.tsv` (8 rows). Module eigengenes per primary sample.

Table S11 `Table_S11_wgcna_module_trait_correlations.tsv` (63 rows). Module eigengene vs trait Pearson correlation, Student p and BH FDR.

Table S12 `Table_S12_go_enrichment_deg_sets.tsv` (3,201 rows). GO over-representation (hypergeometric, BH FDR) for up and down DEG sets and the shared response.

Table S13 `Table_S13_go_enrichment_wgcna_modules.tsv` (8,759 rows). GO over-representation for each WGCNA module.

Table S14 `Table_S14_classification_loocv_predictions.tsv` (272 rows). Leave-one-out held-out predictions and class probabilities (TabPFN, logistic regression, random forest).

Table S15 `Table_S15_classification_metrics.tsv` (32 rows). LOOCV accuracy, balanced accuracy, AUROC and label-permutation p-values per model, task and feature count.

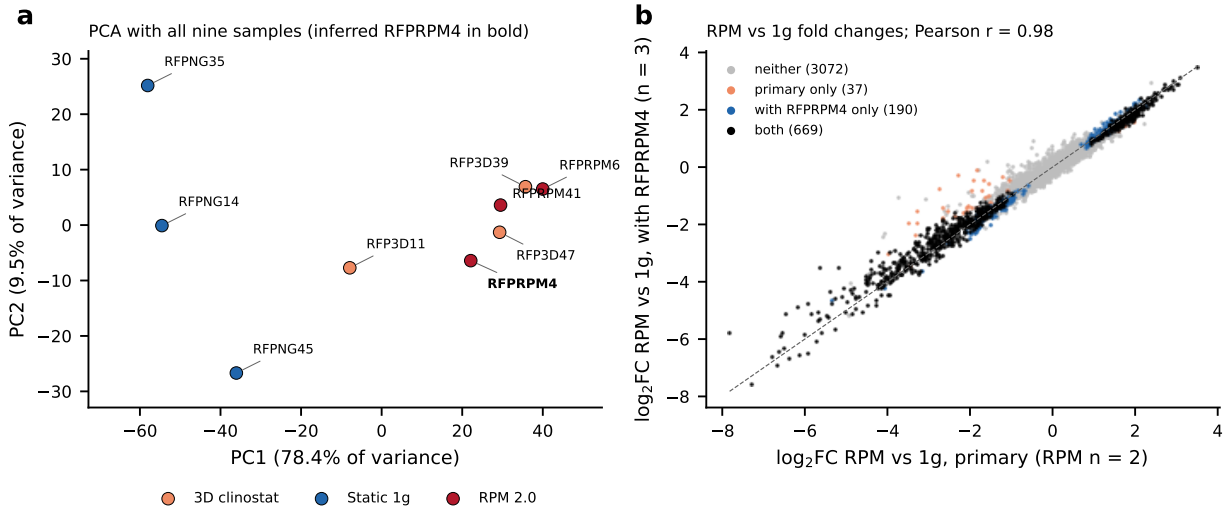
Table S16 `Table_S16_pan_microbial_osdr_compendium_102studies.tsv` (102 rows). Microbial spaceflight and simulated-microgravity studies in NASA OSDR, from the OSDR API.

Table S17 `Table_S17_kegg_mapped_reactions.tsv` (28 rows). Curated reactions with their KEGG-supported M. marinum genes and per-contrast expression changes.

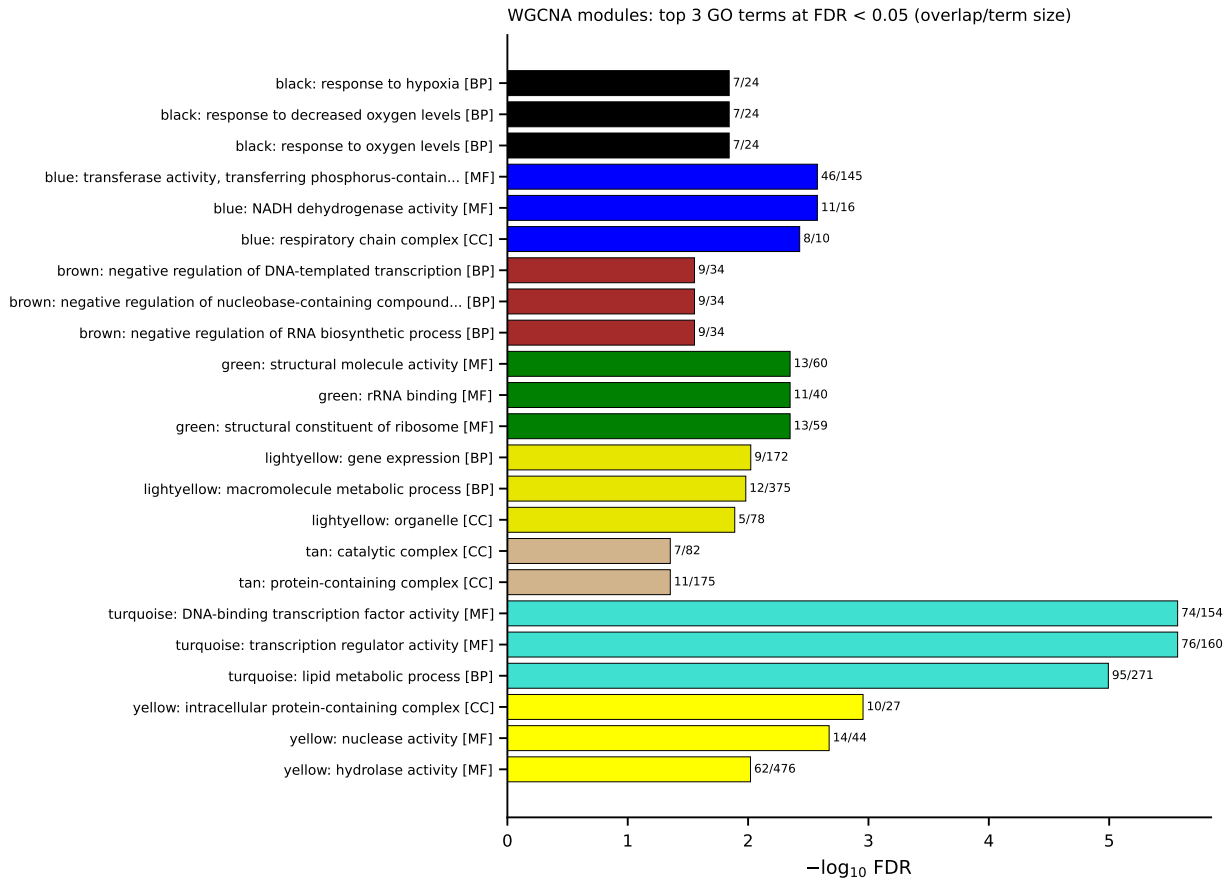
Table S18 `Table_S18_kegg_subsystem_summary.tsv` (16 rows). Per-subsystem, per-contrast summaries of the KEGG-mapped reactions.

Table S19 `Table_S19_metabolic_model.json` (JSON). Reaction model with KEGG-supported gene links (JSON).

Supplementary figures



Supplementary Figure S1. Sensitivity analysis with the inferred RFPRPM4 library. a, PCA of all 9 samples, including the inferred RFPRPM4 library (bold). **b**, RPM vs static 1g fold changes with and without RFPRPM4 (Pearson $r = 0.98$); colours show where each gene is a DEG. Including RFPRPM4 gives 859 RPM DEGs, compared with 706 in the primary analysis.



Supplementary Figure S2. GO enrichment of the WGCNA modules. Up to three GO terms per module at BH FDR < 0.05 (hypergeometric test against all network genes with annotation); labels give genes in the module / annotated genes in the term.